

PES UNIVERSITY
Department of Computer Science and Engineering

Software Engineering - Lab 1

Problem Statement #06:
Alumni Mentorship & Mock Interview Platform
Use Case TC-02: Double-Booking Prevention

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1. Problem Context and Overview

Connecting students with alumni working in specialized domains requires an intelligent matching mechanism, an availability calendar booking system, and structured post-interview scorecard recording.

Target Stakeholders / Actors: Student Mentee, Alumni Mentor

2. Requirements Table

2.1 Functional Requirements (FR-001 to FR-005)

FR ID	Description	Priority	Acceptance Criteria (Pass / Fail)	Rationale
FR-001	The system shall recommend alumni mentors to students based on domain matching (e.g., Embedded Systems, Cloud Computing) and facilitate one-click session booking.	High	Pass: Mentorship session is added to both calendars with a meeting link. Fail: Double-booking of mentor slots occurs.	Reduces manual search effort for students and ensures matches align with relevant domain expertise, improving mentorship quality and engagement.
FR-002	The system shall atomically verify and lock a mentor's selected time slot at the moment of booking confirmation, rejecting any concurrent booking request for the same slot.	High	Pass: When two students attempt to book the same slot simultaneously, only the first request succeeds; the second receives a slot no longer available message with alternative slot suggestions. Fail: Both requests succeed, creating a scheduling conflict (double-booking).	Directly addresses the concurrency risk inherent in a shared mentor calendar; prevents scheduling conflicts that erode mentor trust and platform credibility.

FR-003	The system shall allow alumni mentors to record a structured post-interview scorecard (technical rating, communication rating, and free-text feedback) immediately after a mock interview session concludes.	Medium	Pass: Scorecard is saved, linked to the correct session, and visible to the student within 5 minutes of submission. Fail: Scorecard data is lost, mismatched to the wrong session, or not visible to the student.	Structured, timely feedback is the core value delivered to students; accurate linkage ensures trustworthy and actionable mentorship outcomes.
FR-004	The system shall allow alumni mentors to define, publish, and update recurring or one-off available time slots, which are reflected in real time to students during booking.	Medium	Pass: A slot added or removed by the mentor is immediately reflected as available/unavailable in the student's booking view. Fail: Stale availability data allows booking of a blocked, removed, or already-passed slot.	Accurate, real-time availability is the foundation that mentor matching, booking, and double-booking prevention (FR-002) all depend on.
FR-005	The system shall send automated confirmation notifications at booking time and reminder notifications to both student and mentor prior to the scheduled session.	Low	Pass: Both parties receive a confirmation immediately after booking and a reminder at least 30 minutes before the session. Fail: A notification is missing, significantly delayed, or sent to the wrong party.	Reduces no-shows and reinforces the reliability and professionalism of the overall mentorship experience.

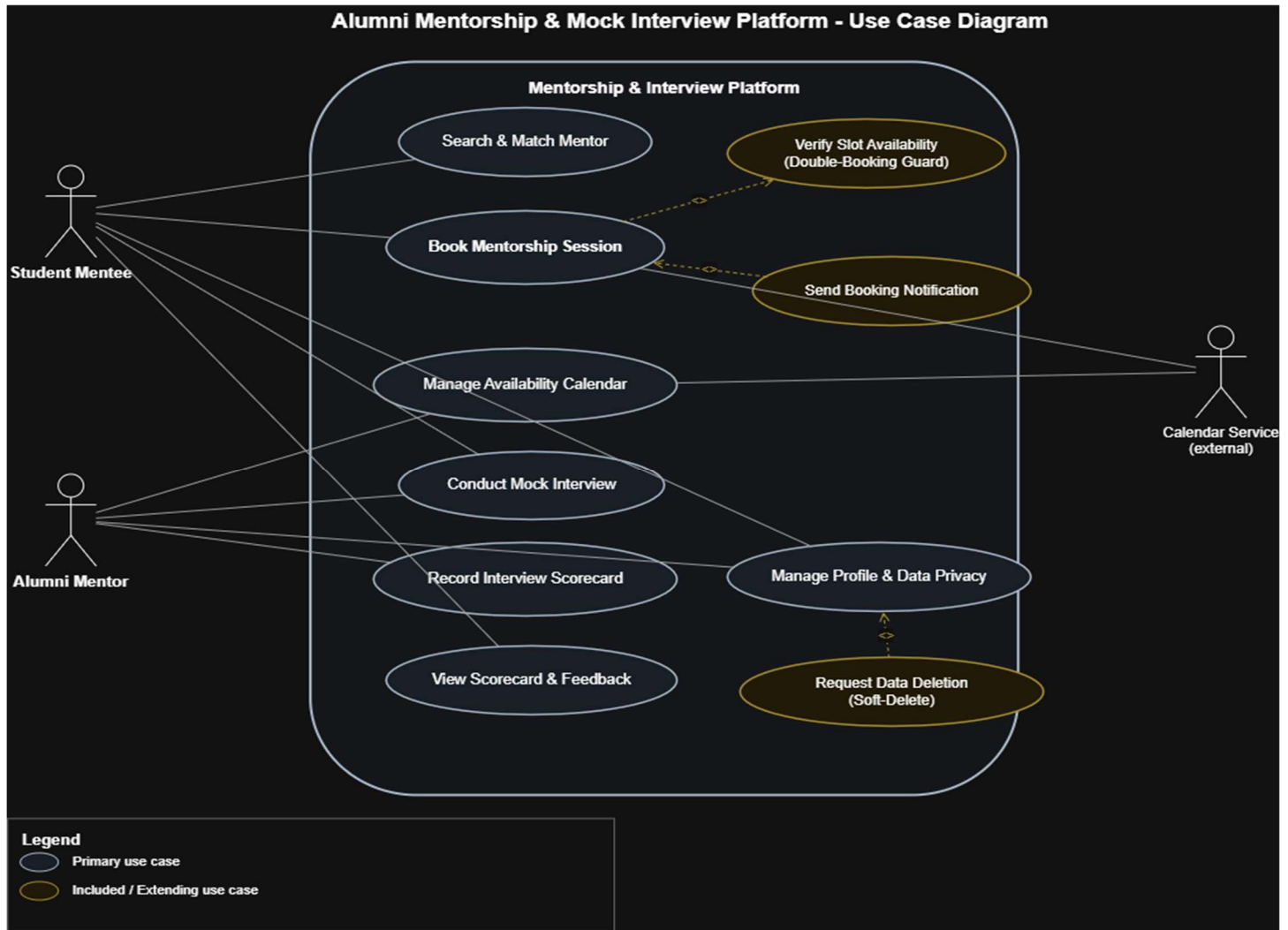
2.2 Non-Functional Requirements (NFR-001 and NFR-002)

NFR ID	Type	Description	Acceptance Criteria (Pass / Fail)	Rationale
NFR-001	Performance & Security	All user profile data and feedback scorecards shall adhere to privacy guidelines and support soft-deletion within 24 hours of a deletion request.	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load, and deletion requests are honored (data marked inaccessible) within 24 hours. Fail: Latency/security benchmarks are missed, or deleted data remains accessible past 24 hours.	Compliance with data-protection expectations builds user trust and mitigates legal and reputational risk from mishandled personal data.

NFR ID	Type	Description	Acceptance Criteria (Pass / Fail)	Rationale
NFR-002	Reliability & Scalability	The booking subsystem shall use atomic, transaction-safe slot-locking so that concurrent booking requests (up to 500 simultaneous requests) are serialized without data corruption, while maintaining 99.9% monthly uptime for the booking service.	Pass: Load testing with 500 concurrent booking attempts on the same slot results in exactly one successful booking and zero data corruption or deadlocks. Fail: More than one booking succeeds for the same slot, or the system deadlocks/crashes under load.	Double-booking prevention (FR-002) is a correctness-critical, concurrency-sensitive operation; explicit, testable reliability guarantees ensure the business rule holds under real-world simultaneous access, not just isolated tests.

3. UML Use-Case Diagram

The diagram below shows both actors and the primary use cases of the platform, along with the required relationships: an <<include>> dependency (Book Mentorship Session includes Verify Slot Availability, which is the mechanism behind double-booking prevention) and <<extend>> dependencies (Send Booking Notification extends Book Mentorship Session; Request Data Deletion extends Manage Profile & Data Privacy).



4. Use-Case Flow Specification - UC-02

Book Mentorship Session (with Double-Booking Prevention)

Use Case Information

Field	Details
Use Case ID	UC-02
Use Case Name	Book Mentorship Session (with Double-Booking Prevention)
Primary Actor	Student Mentee
Supporting Actors	Alumni Mentor, Calendar Service
Related Requirement	FR-002 - Double-Booking Prevention
Includes	Verify Slot Availability
Extended By	Send Booking Notification

1. Preconditions

1. Student Mentee is logged in with an active account.
2. Student has selected an Alumni Mentor.
3. Mentor has an available time slot.
4. Calendar Service is available.

2. Postconditions

Success:

- One confirmed session is created for the selected slot.
- The slot is marked unavailable to other students.
- A calendar event and meeting link are added for both parties.

Failure:

- No second session is created.
- The rejected student's calendar remains unchanged.

3. Main Success Scenario

1. Student opens the mentor's profile and selects Book Session.
2. The system displays the mentor's available slots.
3. Student selects a slot and confirms the booking.
4. The system verifies and locks the selected slot.
5. If available, the system creates the session and marks the slot as booked.
6. The Calendar Service creates the calendar event and meeting link.
7. The system sends a booking confirmation to the student and mentor.
8. The system displays the booking confirmation to the student.

4. Alternate Flow - Double-Booking Attempt (TC-02)

1. Student A and Student B try to book the same slot at nearly the same time.
2. The system locks the slot so only one request can succeed.
3. Student A's booking succeeds.
4. Student B's availability check finds that the slot is already booked.
5. Student B's booking is rejected and no duplicate session is created.
6. Student B sees a "Slot no longer available" message and can choose another slot.

5. Successful Outcome

The selected mentor slot is booked by only one student. Any concurrent booking attempt for the same slot is rejected, preventing double-booking.